

Milestone 1 - Module 1: Web Scraping & Raw Data Collection

1. Methodologies

The methodology for Module 1 focuses on **automated, scalable, and reliable web scraping** of global military data from *GlobalFirepower.com*.

1.1 Data Source Identification

- The primary data source is **GlobalFirepower.com**, a publicly available repository of country-level military, economic, and geographic metrics.
- Metric-specific URLs (55+ pages) were predefined and managed using a configuration-based approach to ensure scalability.

1.2 Base URL Scraping (Power Index)

- The base listing page (countries-listing.php) was scraped separately to extract:
 - Global rank
 - Country full name
 - Country short name
 - Power Index (PwrIndx)
- This separation ensures **modularity** and allows independent validation of ranking data.

1.3 Metric-Level Data Extraction

- Each metric page follows a consistent HTML card structure.
- For every URL:
 1. An HTTP request is sent using a browser-like user agent.
 2. HTML content is parsed using DOM selectors.
 3. Country-level elements are identified dynamically.
 4. Metric values are extracted and mapped to standardized metric names.
- Extracted data is incrementally merged with the base dataset using country names as keys.

1.4 Reliability & Ethical Scraping

- Request throttling (time.sleep(2)) is implemented to avoid server overload.
- HTTP response validation ensures only successful requests are processed.
- Missing or malformed values are safely skipped without breaking execution.
- The system achieves **≥95% URL success rate**.

1.5 Data Storage Strategy

- All extracted data is stored **without transformation** to preserve source fidelity.
- Raw datasets are saved in CSV and Excel formats for downstream processing in Module 2.

2. Libraries Used

Library	Purpose
requests	Sends HTTP requests to GlobalFirepower URLs
BeautifulSoup (bs4)	Parses and navigates HTML content
pandas	Structures scraped data into DataFrames and exports files
re	Extracts URLs from configuration files using regex
time	Controls request intervals to enable responsible scraping
lxml	High-performance HTML parser backend

3. Output

3.1 Generated Datasets

The following raw datasets are produced in Module 1:

File Name	Description
power_index_raw.csv	Country ranks and Power Index data
aircraft-total-fighters_scrape_military_metrics.csv	Sample metric-specific dataset
military_raw_data.csv	Consolidated raw dataset (all metrics)
military_raw_data.xlsx	Excel version for stakeholder review

3.2 Dataset Characteristics

- **Countries Covered:** 145+
- **Metrics Extracted:** 54
- **Categories:**
 - Manpower
 - Air Force
 - Army (Armor)
 - Navy
 - Defense & Economy
 - Energy & Resources
 - Geography

- **Total Records:** ~7,800+
- **Data Type:** Raw, uncleaned, source-level data

3.3 Sample Output Structure

Country_Full_Name	Country_Short_Name	Power_Index	fighter_aircraft	tanks	defense_budget_usd	...
India	IND	0.1023	576	4,614	81B	...
United States	USA	0.0712	1,854	5,500	877B	...

3.4 Intended Usage

- Serves as the **foundation dataset** for:
 - Data cleaning & normalization (Module 2)
 - KPI engineering
 - PowerBI / Tableau dashboards
 - Comparative military analytics

Module 1 Outcome

- Automated scraping pipeline implemented
- 140+ countries successfully covered
- Structured raw dataset generated
- Ready for data preprocessing and visualization